

SARC: STRUCTURED ANALYTIC-RESIDUAL CORRECTION FOR LIGHTWEIGHT MULTICHANNEL SPEECH ENHANCEMENT

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ABSTRACT

Analytic spatial filters preserve a calibrated target response but can leave interference which is difficult to identify from their output alone. We propose Structured Analytic-Residual Correction (SARC), a lightweight framework that combines analytic spatial filtering with neural correction for multichannel speech enhancement. A learned channel-weighting matrix first forms a unit-response analytic estimate from RTF-aligned microphone signals. The key idea is to subtract this estimate from the aligned channels, canceling the modeled target while retaining channel-to-channel nuisance differences. These target-canceling residuals, together with the analytic estimate and a residual scale, guide a compact neural corrector to suppress the remaining interference. On a fixed four-microphone evaluation set, the 46,834-parameter SARC achieves 17.527 dB SI-SDR improvement and 2.730 PESQ.

Index Terms— Multichannel speech enhancement, relative transfer function, neural beamforming, residual correction

1. INTRODUCTION

A compact microphone array captures desired speech and nonstationary noise through multiple microphone paths. When a calibrated target response is available, classical methods can exploit this spatial structure explicitly. The generalized sidelobe canceller (GSC) separates a constrained target path from adaptive cancellation [1, 2], the speech-distortion-weighted multichannel Wiener filter (SDW-MWF) controls target distortion [3], and the maximum-likelihood-based adaptive interference canceller (ML-AIC) combines blocking with a time-varying Gaussian source model [4]. However, such model-based adaptation can be restrictive under rapidly changing interference.

Neural systems learn spatial statistics [5], multichannel filters [6], or direct spectral-spatial mappings [7]. An

unconstrained direct mapping, however, does not explicitly preserve the calibrated target response. Recent hybrid approaches therefore combine learned modules with explicit spatial structure. AFnet uses relative transfer function (RTF)-supervised alignment [8], while other methods embed analytic beamforming or Wiener filtering between learned modules [9, 10, 11]. However, a post-neural-filter that receives only the analytic output cannot use multichannel differences to identify residual interference.

In this paper, we propose Structured Analytic-Residual Correction (SARC), a lightweight multichannel speech enhancement method that retains these spatial differences for neural correction. A pre-calibrated frontal RTF first aligns the microphone signals, and learned channel weighting forms a unit-response analytic estimate z . The key step constructs the spatial residual $\mathbf{r} = \mathbf{y} - \mathbf{1}z$, which cancels the modeled target component. A compact neural corrector then uses $(z, \mathbf{r}, v)$ to estimate the remaining interference and refine z . With only 46,834 parameters, SARC achieves $17.527 \pm .053$ dB SI-SDR improvement, exceeding a matched direct mapping (14.198 dB) and ML-AIC (13.403 dB).

2. PROBLEM FORMULATION

We consider multichannel speech enhancement with a fixed, pre-calibrated target relative transfer function (RTF). Let $\mathbf{x}_{ft} = [x_{1ft}, \dots, x_{Mft}]^T \in \mathbb{C}^M$ denote the array short-time Fourier transform (STFT) at frequency bin f and frame t . For an M -microphone array,

$$\mathbf{x}_{ft} = \mathbf{a}_f s_{ft} + \mathbf{n}_{ft}, \quad (1)$$

where $s_{ft} \in \mathbb{C}$ is the target image at the reference microphone, $\mathbf{a}_f \in \mathbb{C}^M$ is the target RTF, and $\mathbf{n}_{ft}$ contains all other components. We assume that $\mathbf{a}_f$ is fixed and pre-calibrated, with $a_{f,\text{ref}} = 1$. The goal is to estimate s_{ft} from the multichannel observation $\mathbf{x}_{ft}$.

Suppressing (f, t) for clarity, we first canonicalize the microphone signals using the known RTF:

$$\mathbf{y} = D(\mathbf{a})^{-1} \mathbf{x} = \mathbf{1}s + \boldsymbol{\eta}, \quad \boldsymbol{\eta} = D(\mathbf{a})^{-1} \mathbf{n}, \quad (2)$$

where $D(\mathbf{a})$ places the RTF on its diagonal and $\mathbf{1} \in \mathbb{C}^M$ is the all-one vector. After canonicalization, the modeled target

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The code is available at <https://github.com/jgr2021/SARC>.

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component is identical across all aligned microphone channels.

Consider a spatial filter $\mathbf{q}$ satisfying the unit-response constraint

$$\mathbf{q}^H \mathbf{1} = 1. \quad (3)$$

Its output is

$$z = \mathbf{q}^H \mathbf{y} = s + e, \quad e = \mathbf{q}^H \boldsymbol{\eta}, \quad (4)$$

where e denotes the interference remaining in the analytic estimate. The desired enhanced output can be written as

$$\hat{s} = z + \delta, \quad \delta \approx s - z = -e. \quad (5)$$

However, observing only $z = s + e$ provides limited information for distinguishing the target from the residual interference. In contrast, the aligned multichannel observation $\mathbf{y}$ still contains channel-to-channel differences related to the nuisance components. The remaining problem is therefore to construct an observable multichannel reference from $\mathbf{y}$ and z that exposes information about e and can guide the correction δ .

3. STRUCTURED ANALYTIC-RESIDUAL CORRECTION

Figure 1 shows SARC: a precision network determines channel weights, a deterministic layer forms $(z, \mathbf{r}, v)$, and a correction network predicts an update to cancel the remaining interference.

3.1. RTF canonicalization

Equation (2) divides out the pre-calibrated frontal RTF, making the target response the fixed direction $\mathbf{1}$ [13, 8]. No microphone coordinates are used; spatial responses enter through $\mathbf{a}_f$.

3.2. Precision network and analytic estimate

To infer relative channel reliability, we normalize $\mathbf{y}$ by its array root-mean-square (RMS) ρ . Each $u_m = y_m/\rho$ contributes real, imaginary, and log-magnitude feature maps [14, 15]; common phase, pairwise power balance, and normalized channel spread add four global feature maps, giving $3M+4 = 16$ inputs.

The analytic estimator requires a positive-definite weighting matrix for channel combination and residual measurement. Neural Cholesky parameterization has likewise been used to guarantee valid covariance estimates [16]. Here four time-causal convolutional blocks and a 48-unit forward GRU output the M^2 real degrees of freedom of a complex Cholesky factor:

$$\begin{aligned} L_{mm} &= \exp(1.5 \tanh \alpha_m), \\ L_{mk} &= .7(\tanh \beta_{mk}^R + j \tanh \beta_{mk}^I), \quad m > k. \end{aligned} \quad (6)$$

Here α_m , β_{mk}^R , and β_{mk}^I are real network outputs. For $M = 4$, four positive diagonal and six complex lower-triangular entries require 16 real values. The learned nuisance weighting matrix $\boldsymbol{\Psi} = \mathbf{L}\mathbf{L}^H \succ \mathbf{0}$ is positive definite by construction. A fixed MVDR-form unit-response estimator [12] then computes

$$d = \mathbf{1}^H \boldsymbol{\Psi} \mathbf{1}, \quad \mathbf{q} = \frac{\boldsymbol{\Psi} \mathbf{1}}{d}, \quad z = \mathbf{q}^H \mathbf{y}. \quad (7)$$

Because $\mathbf{q}^H \mathbf{1} = 1$, modeled target speech passes through z unchanged; equivalently, $\mathbf{w} = \mathbf{D}(\mathbf{a})^{-H} \mathbf{q}$ satisfies $\mathbf{w}^H \mathbf{a} = 1$ in the microphone domain.

3.3. Spatial residual as an interference reference

Using (??), the deterministic layer forms

$$\mathbf{r} = \mathbf{y} - \mathbf{1}z = \boldsymbol{\eta} - \mathbf{1}e, \quad \mathbf{q}^H \mathbf{r} = 0. \quad (8)$$

The modeled target cancels in $\mathbf{r}$, leaving spatial nuisance references for estimating e , as in GSC and ML-AIC blocking paths [1, 4]. The raw pair also reconstructs $\mathbf{y} = \mathbf{1}z + \mathbf{r}$ before feature normalization; this is an auxiliary structural property.

The corrector also needs a scale for the plausible error of z . We define the scale proxy

$$v = \frac{\mathbf{r}^H \boldsymbol{\Psi} \mathbf{r}}{(M-1)d} \quad (9)$$

It converts precision-weighted disagreement to the scale of z . The quantity v summarizes residual energy to condition the corrector.

3.4. Correction network

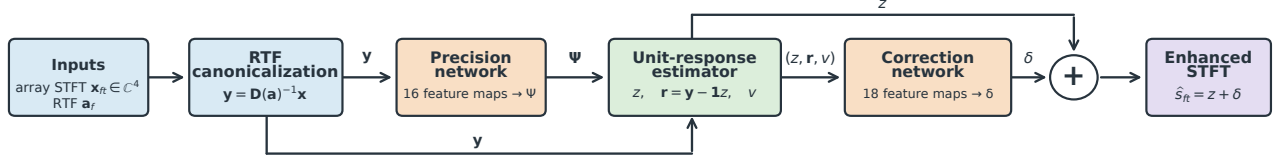
The corrector receives six bounded feature maps describing the magnitude and phase of z and the absolute and relative scales of v . Each RMS-normalized residual channel contributes real, imaginary, and log-magnitude feature maps, giving $6 + 3M = 18$ inputs. Thus z supplies the current speech estimate, $\mathbf{r}$ supplies spatial interference references, and v supplies a conditioning scale.

After four causal convolutional blocks, a 24-unit bidirectional frequency GRU and a 48-unit forward temporal GRU follow the frequency-time factorization of dual-path enhancement models [15]. They output two real coordinates h_R and h_I for Cartesian complex prediction [14]. The bounded correction is

$$\begin{aligned} \delta &= C|z| [\tanh(h_R) + j \tanh(h_I)], \quad C = 4, \\ \hat{s} &= z + \delta. \end{aligned} \quad (10)$$

Here $|z|$ supplies the local STFT amplitude scale, while $\tanh(\cdot)$ bounds each Cartesian update by $C|z|$; v conditions the network but does not scale its output. The supervised update $s - z = -e$ trains the corrector to cancel the interference in z . The precision and correction networks contain 18,768 and 28,066 parameters, respectively.

(a) SARC signal flow



(b) Learned modules

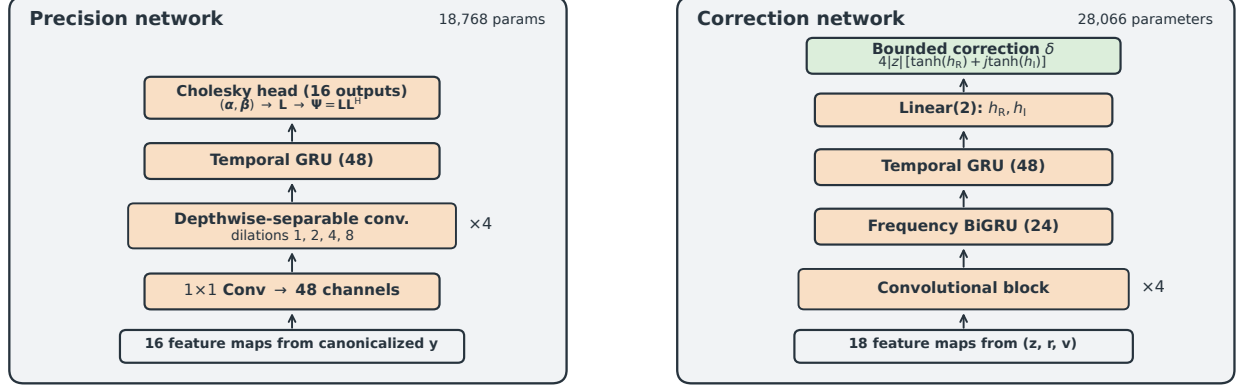


Fig. 1. SARC architecture. Blue blocks denote observed inputs or fixed calibration operations, orange blocks are learned, green blocks are deterministic, and gray blocks construct features. The precision network outputs $\mathbf{L}$; the structured front end forms Ψ , z , v , and $\mathbf{r}$. The residual exposes channel-to-channel nuisance differences for neural correction, while z bypasses the corrector so that $\hat{s} = z + \delta$.

3.5. Training objective

Let $S = [s_{ft}]$ and $\hat{S} = [\hat{s}_{ft}]$ be the target and predicted STFTs, s_{wav} the clean waveform, and $\langle \cdot \rangle$ the mean over valid bins and examples. With power-law compression $c(X) = |X|^{0.3} e^{j\angle X}$ [17], we combine spectral error and SI-SDR [18]:

$$\begin{aligned} \mathcal{L}_C(X) &= \frac{\langle |c(X) - c(S)|^2 \rangle}{\langle |c(S)|^2 \rangle}, \\ \mathcal{L}_{\text{SI}}(X) &= -\frac{1}{20} \text{SI-SDR}(\text{iSTFT}(X), s_{\text{wav}}), \\ \mathcal{L}_{\text{enh}}(X; \lambda) &= \mathcal{L}_C(X) + \lambda \mathcal{L}_{\text{SI}}(X). \end{aligned} \quad (11)$$

During training, normalize the known nuisance $\boldsymbol{\eta} = \mathbf{y} - \mathbf{1}S$ as $\tilde{\boldsymbol{\eta}} = \boldsymbol{\eta} / (M^{-1} \sum_m |\eta_m|^2)^{1/2}$. Motivated by the quadratic and log-determinant terms of Gaussian precision fitting [19], we introduce the regularizer

$$\mathcal{L}_{\text{prec}} = \left\langle \frac{\tilde{\boldsymbol{\eta}}^H \Psi \tilde{\boldsymbol{\eta}} - \log \det \Psi}{M} \right\rangle \quad (12)$$

to fit relative nuisance structure and fix the scale of Ψ without asserting an exact inverse covariance. Let $\bar{p}_s =$

$F^{-1} \sum_f |S_{ft}|^2$. The auxiliary losses are

$$\begin{aligned} \mathcal{L}_{\text{corr}} &= \left\langle \frac{|\delta - (S - z)|^2}{v + 10^{-3} \bar{p}_s} \right\rangle, \\ \mathcal{L}_{\text{size}} &= \left\langle \frac{|\delta|^2}{|z|^2 + v} \right\rangle, \\ \mathcal{L}_{\text{sc}} &= \left\langle \frac{|z - S|^2}{v} + \log \frac{v}{\bar{p}_s} \right\rangle. \end{aligned} \quad (13)$$

Motivated by staged alignment/filter optimization [8], SARC first trains the analytic estimator, then the corrector with the precision network frozen, and finally both modules jointly:

$$\begin{aligned} \mathcal{L}_{\text{sp}} &= \mathcal{L}_{\text{enh}}(z; .10) + .02 \mathcal{L}_{\text{prec}} + .001 \langle \|\mathbf{q}\|_2^2 \rangle, \\ \mathcal{L}_{\text{net}} &= \mathcal{L}_{\text{enh}}(\hat{S}; .12) + .03 \mathcal{L}_{\text{corr}} + .001 \mathcal{L}_{\text{size}}, \\ \mathcal{L}_{\text{joint}} &= \mathcal{L}_{\text{net}} + .10 \mathcal{L}_C(z) + .02 \mathcal{L}_{\text{prec}} + .01 \mathcal{L}_{\text{sc}}. \end{aligned} \quad (14)$$

4. EXPERIMENTS

4.1. Experimental setup

All systems use the same four microphone channels at 16 kHz, with the first selected channel as reference. The fixed evaluation set has 80 mixtures from 20 unseen LibriSpeech speakers [20], balanced over four evaluation conditions and five SNRs from -10 to 10 dB, with one or two

Table 1. Comparison on the fixed four-microphone evaluation set. Learned quality scores report mean $\pm$ sample standard deviation over three runs. Runtimes are medians for a 4-s input (Sec. 4.2); dashes denote no timing test.

System	Mics	GPU (ms)	CPU (ms)	Parameters	Δ SI-SDR (dB)	STOI	PESQ
Input microphone	1	—	—	0	0	.766	1.237
Norm-constrained GSC [2]	4	—	174.0	0	7.854	.922	1.583
Online SDW-MWF [3]	4	—	348.9	0	8.138	.874	1.593
ML-AIC [4]	4	—	209.6	0	13.403	.988	2.288
Online McNet [7]	4	170.0	2,831.5	1,837,250	$11.355 \pm .700$	$.915 \pm .008$	$2.263 \pm .071$
AFnet [8]	4	74.6	1,459.0	2,554,896	$10.102 \pm .018$	$.923 \pm .005$	$2.257 \pm .059$
EaBNet [6]	4	37.3	701.6	2,825,160	$12.105 \pm .133$	$.933 \pm .001$	$2.142 \pm .062$
SARC	4	28.0	458.7	46,834	$17.527 \pm .053$	$.961 \pm .001$	$2.730 \pm .057$

source-disjoint ESC-50 interferers [21]. The spatial transfer functions are shared across compared systems. Training, validation, and test crops use disjoint time intervals.

RTF-dependent methods receive the same fixed frontal calibration, and all neural baselines receive the same four signals. SARC uses a 512-point STFT, a 25-ms square-root Hann window, and a 10-ms hop. We report Δ SI-SDR, STOI [22], and wideband PESQ [23]. Learned systems report mean and sample standard deviation over three runs.

GSC, SDW-MWF, and ML-AIC follow their published algorithms. McNet uses author code and an adapted checkpoint, EaBNet uses the official architecture trained locally, and AFnet is reimplemented from its paper. Outputs are rescored on identical trials. Paired 95% confidence intervals use 30,000-draw cluster bootstraps over seed, speaker, and evaluation condition.

4.2. Comparison with existing systems

SARC obtains the highest mean Δ SI-SDR and PESQ. Against ML-AIC, it improves SI-SDR by 4.125 dB (95% CI: 2.109–5.941 dB) and PESQ by .441 (95% CI: .208–.643). ML-AIC has the highest STOI (.988 versus .961), with a .0267 advantage (95% CI: .0142–.0433). Among the neural baselines, EaBNet gives the highest Δ SI-SDR at 12.105 dB; SARC exceeds it by 5.423 dB (95% CI: 3.630–6.959 dB).

Figure 2 shows a 0-dB example selected near the median SARC improvement in this SNR group. SARC suppresses background energy while retaining visible harmonic structure.

Table 1 reports full-utterance times on an RTX 4060 laptop GPU and single-thread i7-13650HX CPU, including STFT/iSTFT but excluding I/O and device transfers. Neural timings pool two rounds of 10 runs with reversed method order and 3 warmups per round (seed 1, batch 1, PyTorch 2.5.1, FP32). Classical timings retain 10 runs from a separate session (MATLAB R2024b, FP64).

In this benchmark, SARC is 1.3–6.1 times faster on GPU and 1.5–6.2 times faster on CPU than the neural baselines. Classical implementations show lower CPU times, although

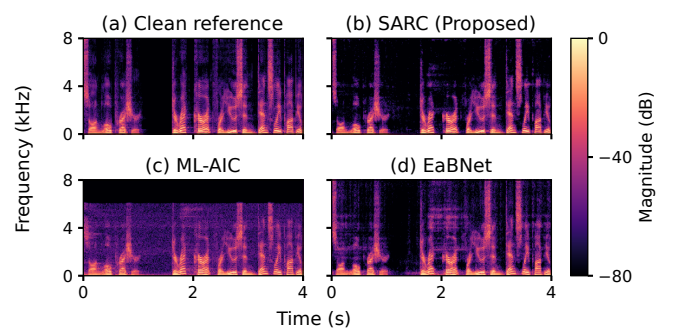


Fig. 2. Synchronized clean reference and enhanced STFT magnitudes, with a shared color scale and no per-signal normalization.

Table 2. Ablations of the SARC mechanism on the same evaluation set.

Configuration	Δ SI-SDR (dB)	Component examined
Analytic z only	$11.333 \pm .276$	Without correction network
Identity precision + correction	$13.798 \pm .230$	Without channel weighting
Direct mapping	$14.198 \pm .543$	Without analytic structure
ML-AIC + same correction	$15.301 \pm .749$	Different analytic front-end
SARC	$17.527 \pm .053$	Proposed method

cross-family comparisons involve different sessions, software, and precision.

4.3. Ablation study

Table 2 supports four design choices: correction improves the analytic estimate; learned channel weighting improves on identity precision; analytic structure outperforms capacity-matched direct mapping; and SARC’s front end performs better than ML-AIC with the same correction topology. These controls support the combined design, but do not isolate residual-input benefits.

5. CONCLUSION

SARC uses target-canceling spatial residuals to guide a bounded neural correction of interference remaining in a unit-response analytic estimate. On the fixed four-microphone evaluation set, the 46,834-parameter model achieves $17.527 \pm .053$ dB SI-SDR improvement and $2.730 \pm .057$ PESQ, exceeding the matched direct mapping and ML-AIC in these metrics; ML-AIC retains the highest STOI. The results support combining calibrated spatial estimation with residual-guided correction.

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